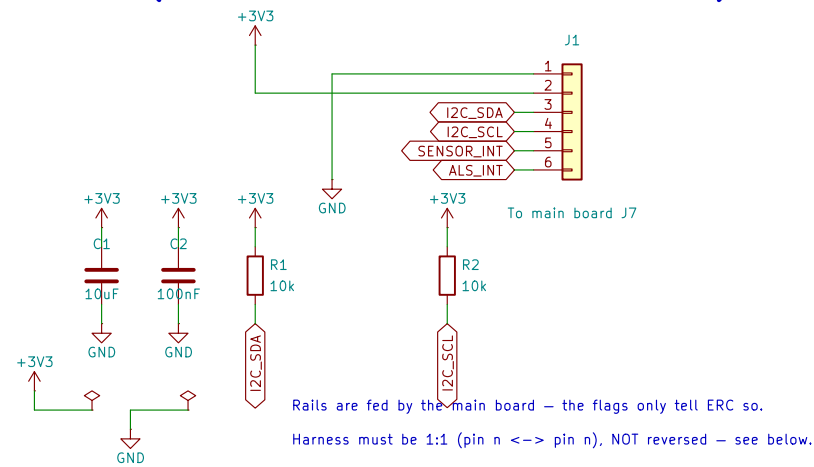


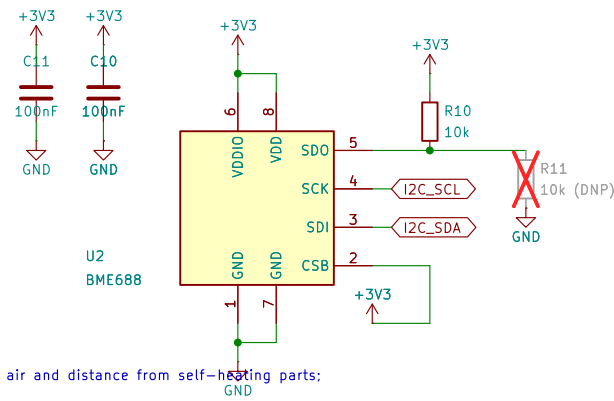
HOST LINK (J1 -> main-board J7, JST ZH 1x06)



HARNESS POLARITY IS A DESTRUCTIVE FAULT

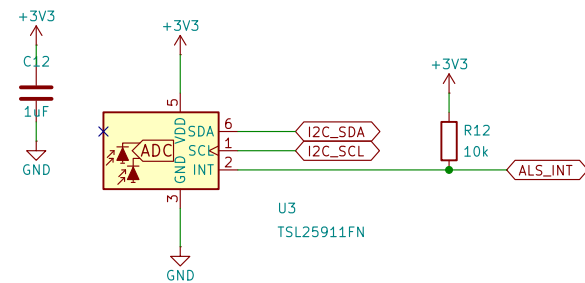
Both ends are ZHR-6 and J1 mirrors J7 1:1, so a cable loaded BACKWARDS plugs in perfectly – and J1.5 then receives J7.2 +3V3 while this board's VDDIO is 0 V. BNO08X abs max (Fig. 6-1) is "VDDIO + 0.3 V at any logic pin", since-limited only by the main board's R97 10k; assume U1 is destroyed. J10 (knob) is the SAME ZX 10x6 header 13.5 mm from J7, with +5 V on pin 2 – that mis-mate plugs 5 V on +3V3 and takes all three sensors at once. Label both cables; check before first power-up.

ENVIRONMENT – BME688 (I2C 0x77)



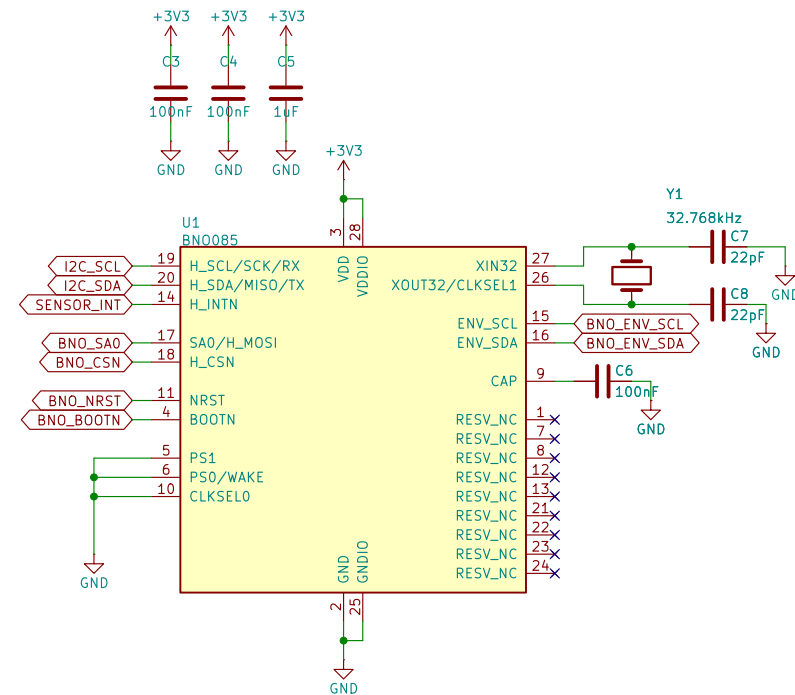
Gas sensor: needs ambient air and distance from self-heating parts;
keep an enclosure vent over U2 and do not conformal-coat it.

AMBIENT LIGHT – TSL2591 (I2C 0x29)

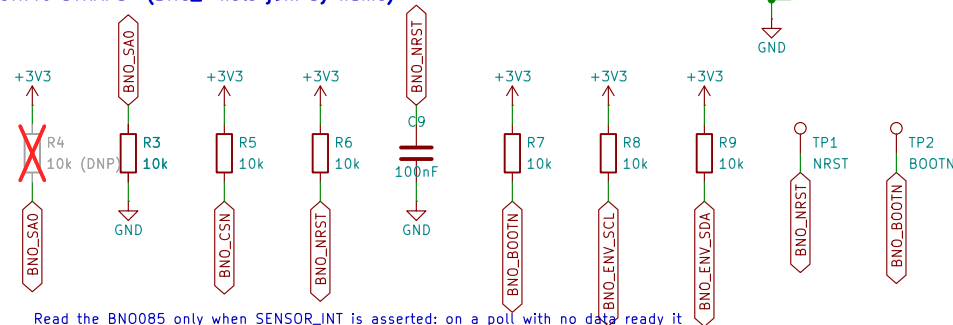


Clear window / no shadow over U3; keep it off-axis from the dial LEDs.

IMU – BN0085 (I2C 0x4A, H_INTN -> SENSOR_INT)



CONFIG STRAPS (BNO_* nets join by name)



Read the BNO085 only when SENSOR_INT is asserted: on a poll with no data ready it STRETCHES SCL (§1.2.2.1) and would stall the whole shared bus (BME688/TSL2591/MCP23017).

NOTES

BOARD Small daughterboard on the cube's front/underside, tied to main-board J7 by a 6-way JST ZH harness (B6B-ZR both ends, 1:1). +3V3 and GND arrive on the cable; there is no regulator here. Tail: JST specs the 3.4 mm variant for a 1.6 mm board, but B6B-ZR-3.4 is made-to-order (MOQ 2000, 16 wk), and 2.7 mm leaves 1.1 mm protruding, which solders fine. Deviation accepted 2026-08-07.

ASSEMBLY All three sensors are LEADLESS (LGA-28 / LGA-8 / DFN-6) – this board CANNOT be hand-soldered with an iron and is the 'path 2b' SMT-assembled daughterboard of datasheet/README.md. Order it fab-assembled (or stencil + hotplate). The main board's hand-solder-only rule does not apply here. J1 is the ONLY through-hole part: it needs a third operation (selective/hand solder) after both reflows – price that before committing, it is usually quoted per board, not per joint.

U2 HANDLING (BME688) A metal-oxide gas hotplate does not fail when mishandled, it reads a wrong VOC baseline forever. MSL 1 (no bake), peak reflow 260 C (\$7.6), and >= 50 um solder height AFTER reflow (\$7.7, for mechanical decoupling – that is a stencil-thickness decision, not a layout one. NO aqueous wash, no flux over the port, no conformal coat, no silicone/siloxane exposure. Bosch's rules are Bosch's HSMI, which \$7.7 defers to: [../datasheet/sensor_env_bme68x_hsmi.pdf](#) (no BME688 – specific HSMI is published; the BME680 one is the same package, lid vent and MOX element).

12C One bus, three addresses: BNO085 0x4A (R3; R4 -> 0x4B) - BME688 0x77 (R10; R11 -> 0x76) - TSL2591 0x29 (fixed). No clash with the main board's MCP23017 (0x20) or the amp (0x62/0x63). Address straps are 10k, not 0R: R3+R4 or R10+R11 fitted TOGETHER would otherwise short the main board's +3V3 to GND. Fit exactly one of each pair, the alternates are flagged DNP. Bus pull-ups: 4.7k on the main board + R1/R2 10k here == 3.2k effective.

IRQ J7.5 SENSOR_INT = BNO085 H_INTN only (push-pull, active low: main-board R97 10k holds it high when the harness is unplugged). J7.6 ALS_INT = TSL2591 INT, open drain, pulled up by R12 HERE and only here – it lands on main-board MCP23017 GPB3, which must enable its internal pull-up (GPPU3 = 1) so that pin does not float when this board is unplugged. See FIRMWARE.md §6.5.

Sheet: / File: sensor.kicad_sch			
Title: Wooden Smart Clock – sensor board			
Size: A3	Date: 2026-08-07	Rev: v0.2	
KiCad E.D.A. 10.0.4	Id: 1/1		